By the way, strictly for those with a professional interest, there is an analogy here with the argument over group selection. We can think of an individual organism as a 'group' of cells. A form of group selection can be made to work, provided some means can be found for increasing the ratio of between-group variation to within-group variation. Bottle-wrack's reproductive habit has exactly the effect of increasing this ratio; splurge-weed's habit has just the opposite effect. There are also similarities, which may be revealing but which I shall not explore, between 'bottlenecking' and two other ideas that have dominated this chapter. Firstly the idea that parasites will cooperate with hosts to the extent that their genes pass to the next generation in the same reproductive cells as the genes of the hosts—squeezing through the same bottleneck. And secondly the idea that the cells of a sexually reproducing body cooperate with each other only because meiosis is scrupulously fair.

To sum up, we have seen three reasons why a bottlenecked life history tends to foster the evolution of the organism as a discrete and unitary vehicle. The three may be labelled, respectively, 'back to the drawing board', 'orderly timing-cycle', and 'cellular uniformity'. Which came first, the bottlenecking of the life cycle, or the discrete organism? I should like to think that they evolved together. Indeed I suspect that the essential, defining feature of an individual organism *is* that it is a unit that begins and ends with a single-celled bottleneck. If life cycles become bottlenecked, living material seems bound to become boxed into discrete, unitary organisms. And the more that living material is boxed into discrete survival machines, the more will the cells of those survival machines concentrate their efforts on that special class of cells that are destined to ferry their shared genes through the bottleneck into the next generation. The two phenomena, bottlenecked life cycles and discrete organisms, go hand in hand. As each evolves, it reinforces the other. The two are mutually enhancing, like the spiralling feelings of a woman and a man during the progress of a love affair.

The Extended Phenotype is a long book and its argument cannot easily be crammed into one chapter. I have been obliged to adopt here a condensed, rather intuitive, even impressionistic style. I hope, nevertheless, that I have succeeded in conveying the flavour of the argument.